

Decision Tree Algorithm

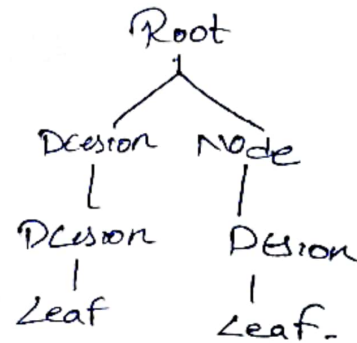
- * It's in Supervised ML.
- * Used for Regression & Classification.

What is DTR

A Decision Tree splits data step by step based on Conditions until it reaches a final Prediction

Structure of DT

- * Root
- * Decision Nodes
- * Leaf Nodes.



Root : Starting Point which has max Information.

Decision nodes : helps to make decision.

Leaf Node : Terminal Point which gives the o/p.

Pruning

Is to prevent overfitting and make the model simpler and more accurate on new data.

(or)

Process of removing (or) Cut down the level of decision nodes

this is usually done to avoid/prevent overfitting.

Pure Split $\div$ Decide a Particular o/p.

Impure Split $\div$ Can't decide a Particular. o/p.

Splitting Criteria for Decision Tree

① Information Gain.

② Gini Impurity.

* Information Gain:

measure How much the feature is free of Impurities.

used to decide which feature gives the best split.

* Calculated by using Entropy.

$\rightarrow$ Measures of Impurity.

Steps of Entropy

① Calculate the Entropy of target:

Play $\begin{cases} \text{yes (9)} \\ \text{No (5)} \end{cases} \Rightarrow$ Probability of Play
 $P(\text{yes}) = 9/14$
 $P(\text{No}) = 5/14$

$$\begin{aligned} H(S) &= -P(\text{yes}) \log(P(\text{yes})) - P(\text{No}) \log(P(\text{No})) \\ &= -P_+ \log P_+ - P_- \log P_- \\ &= -9/14 \log(9/14) - 5/14 \log(5/14) \\ &= 0.92 \end{aligned}$$

Here $\boxed{H = \text{Entropy}}$

② Calculate Entropy for Each feature.

$$H(\text{Sunny}) = -2/5 \log(2/5) - 3/5 \log(3/5) = 0.97$$

$$H(\text{Rainy})^{\frac{3}{5}, \frac{2}{5}} = 0.97.$$

$$H(\text{Cloudy}) = 0$$

$$\begin{aligned} \textcircled{3} \text{ Gain (Information)} &= H(\text{play}) - \left(\sum_i H(\text{feature instance}) \right) \\ &= H(\text{play}) - \frac{SV}{S} H(\text{Rainy}) - \frac{SV}{S} H(\text{Cloudy}) \\ &\quad - \frac{SV}{S} H(\text{Sunny}) \end{aligned}$$

$\therefore S \rightarrow$ Total No. of Sample

$SV \rightarrow$ No. of Instance of that attribute.

Gain

it Calculates Information Gain for Each features and chooses the highest values.

higher values chooses as Root Node.

* Gini Impurity :

$\hookrightarrow$ default Criteria.

* Gini Index should be less.

formula

$$\boxed{\text{Gini} = 1 - \sum P_i^2}$$

Ex

$$\begin{aligned} C_1 = 5 & \quad P(C_1) = 5/5 \\ C_2 = 0 & \quad P(C_2) = 0/5 \end{aligned}$$

$$1 - \sum (5/5)^2 - (0/5)^2$$

$$1 - 1 = 0.$$

* Decision tree for regression we use variance reduction factor. (It should be min)

* for classification we use Probability.

Ensemble Learning Technique

used to combine multiple models to produce a stronger model. Instead of using one model. Several model work together to improve accuracy.

Two type of Techniques

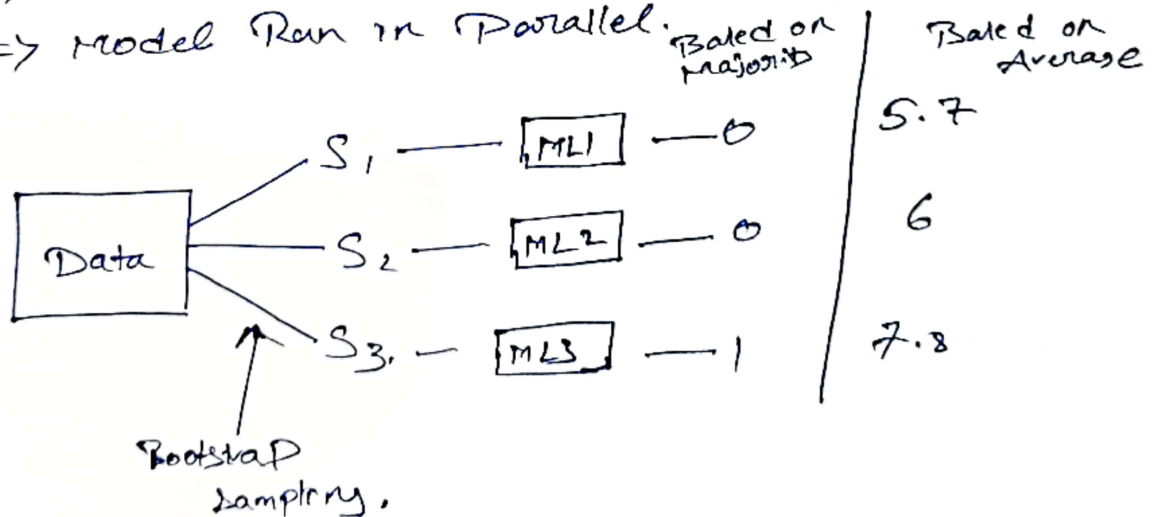
- ① Bagging
- ② Boosting.

Bagging :-

1. Create many random dataset using Bootstrap Sampling.
2. Train many models (usually decision tree).
3. Combine prediction using average/majority vote.

=> Reduce Variance

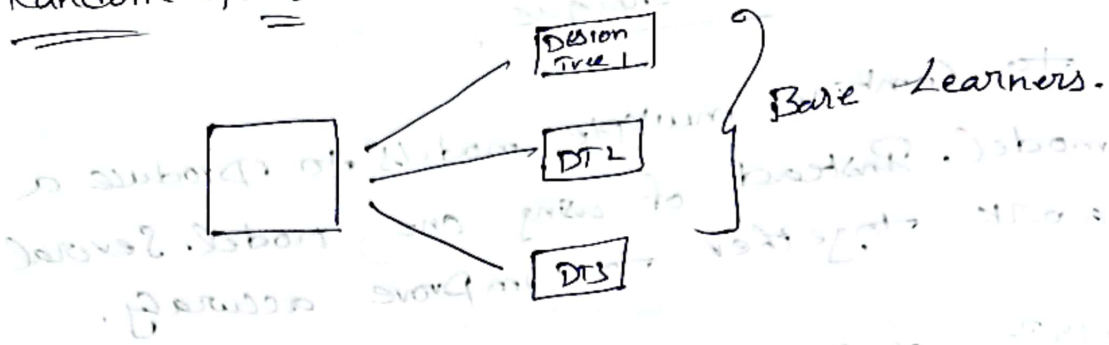
=> Model Run in Parallel.



Bootstrap Sampling

used to create multiple datasets from one original dataset by random sampling with replacement.

Random Forest = Bagging Technique.



Dataset
↓
Bootstrap Sampling
↓
Multiple Decision Tree
↓
Voting & Average
↓
Final Prediction.

(For regression models) Average prediction of all trees.

For classification models, majority class prediction.

Advantages of Random Forest:

- 1. High accuracy
- 2. Robust to overfitting
- 3. Handles noisy data
- 4. Feature importance

Boosting

multiple models are trained sequentially, and each new model tries to correct the mistakes of the previous model.

Types:

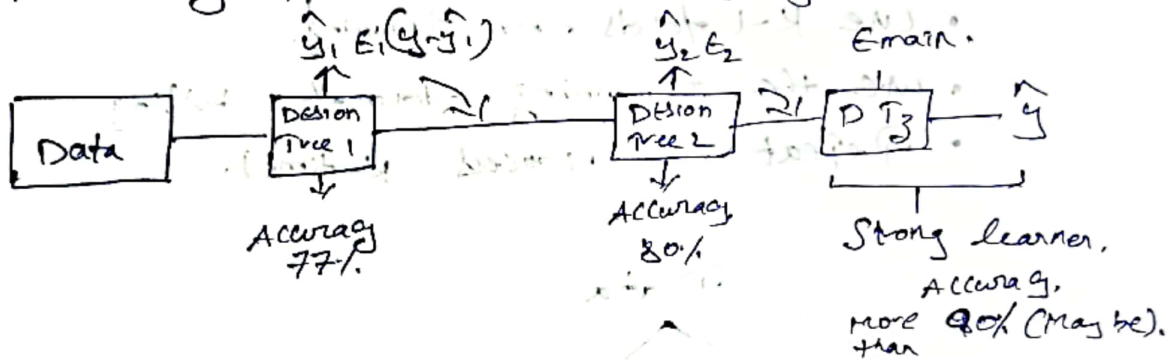
- * Gradient Boosting

- * Xg boosting.

- * Ada Boosting.

- * Gradient Boosting:

Each new model tries to reduce the error made by previous model using gradient descent



- * High Accuracy

- * used in many ML tasks.

- * Xg boosting:

is also known as Extreme gradient Boosting. It is improved and faster version of gradient boosting.

- * Not sensitive to Null values.

- * Hyperparameter & cross validation are inbuilt.

Cross validation

Splitting the dataset multiple times into training and testing parts to evaluate model performance more reliable.

Types

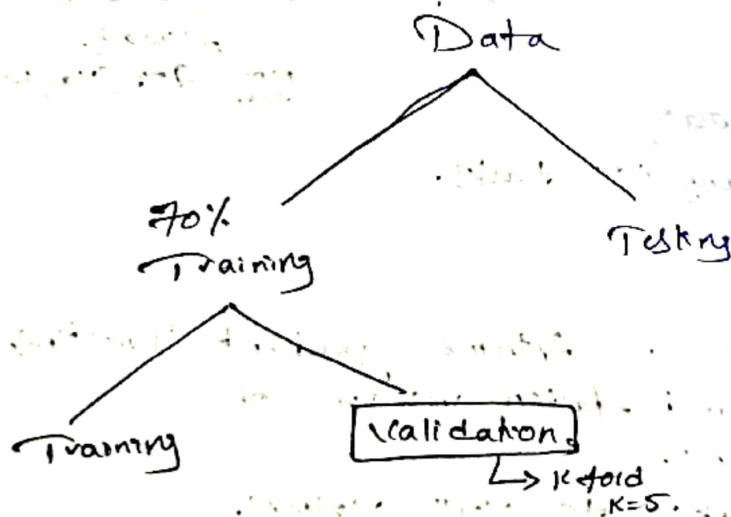
- * K-fold CV
- * Stratified K-fold CV

It is a technique to prevent overfitting and choose best parameter.

* K-fold CV

Instead of splitting the dataset only once, the data is split K times, and the model is trained and tested K times.

- Divide dataset into K Equal Parts.
- Use K-1 folds for training.
- Use the remaining fold for testing.
- Repeat the process K-times.



K-fold Cross Validation

K=5.

EXP 1

T	T	T	T	V
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 → Acc 1

500 data

EXP 2

T	T	T	V	T
---	---	---	---	---

 → Acc 2

Ex 5

V	T	T	T	T
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→ Acc 5.

Bias Variance Trade of

	overfitting	underfitting	good Model
Training	good acc Low bias	Bad acc high bias	good acc Low bias
Testing	High Variance Bad acc	high Variance Bad acc	Low Variance good acc